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Battery field data and why it matters: Foundations for real-world electric vehicles

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ABSTRACT

The traditional paradigm of battery research, primarily rooted in controlled laboratory experiments, is being fundamentally reshaped by the influx of real-world field data. Although laboratory tests remain indispensable for isolating specific electrochemical mechanisms, they fall short of capturing the complex phenomena that arise under practical operating conditions. Field data offers essential insights into this complexity by revealing the intricate interplay among dynamic loads, thermal transients, and path-dependent degradation—interactions often obscured in simplified test protocols. This discrepancy underscores a significant gap in understanding, highlighting that field data is not merely a validation tool, but a vital source for uncovering new physics governing battery performance and aging in realistic environments. Harnessing this potential requires addressing critical challenges—from data quality and privacy to the integration of emerging methodologies in feature engineering, fleet analytics, and physics-informed machine learning. This review surveys large-scale fleet datasets alongside high-resolution vehicle- and cell-level measurements, and examines methodologies spanning state estimation, fault detection, and energy optimization. These developments collectively point to a paradigm shift in battery research—from passive diagnostics toward proactive lifecycle management. Ultimately, this trajectory leads to generalized battery foundation models: continuously evolving digital twins that actively shape, rather than merely predict, a battery's entire lifecycle.

1. Introduction

The global transition to electric vehicles (EVs) is creating a new, data-rich ecosystem. As the EV fleet management market expands – projected to grow from $21.94 billion in 2023 to $29.66 billion by 2028 as shown in Fig. 1(a) – the industry's focus is shifting from hardware-centric capital expenditures to software-driven operational efficiency, a transition reflected in the changing cost structures illustrated in Fig. 1(b). This shift is generating an unprecedented volume of real-world operational data, or "field data", which is fundamentally challenging the traditional, laboratory-centric understanding of battery performance and degradation.

Controlled laboratory experiments, while crucial for isolating specific electrochemical mechanisms, cannot replicate the stochastic and path-dependent aging processes that occur in real-world applications [3–7]. Recent findings from field data analysis have revealed a critical distinction between temporary "apparent aging" and permanent "true irreversible aging" [8]. Conventional accelerated lab tests, often using constant power cycles, can induce reversible degradation effects that lead to an overestimation of the actual aging rate [9,10]. In contrast, the dynamic load profiles and frequent rest periods characteristic of real-world driving have been shown to mitigate these effects and can even enhance battery lifetime compared to static cycling conditions [11]. Moreover, field data possesses an intrinsic structural complexity: the system architecture itself is embedded within the data. Pack-level measurements are inherently influenced by the connection topology, such as the position of the system terminal (ST) and the presence of cross-connectors. These measurements convolve the pure electrochemical response of the cells with the physical resistances of the entire assembly, including terminal (R_{tp}, R_{tn}), busbar (R_{bp}, R_{bn}), inter-cell (R_m), and welding (R_{wp}, R_{wn}) resistances [12]. Therefore, the signals captured from a vehicle are a reflection of the complete, integrated system, not just of isolated cells. This paradigm shift underscores that field data is not merely a supplement to lab data but an essential source of ground truth.

This growing recognition of field data's importance has fueled a dramatic acceleration in research activity. A literature analysis focused

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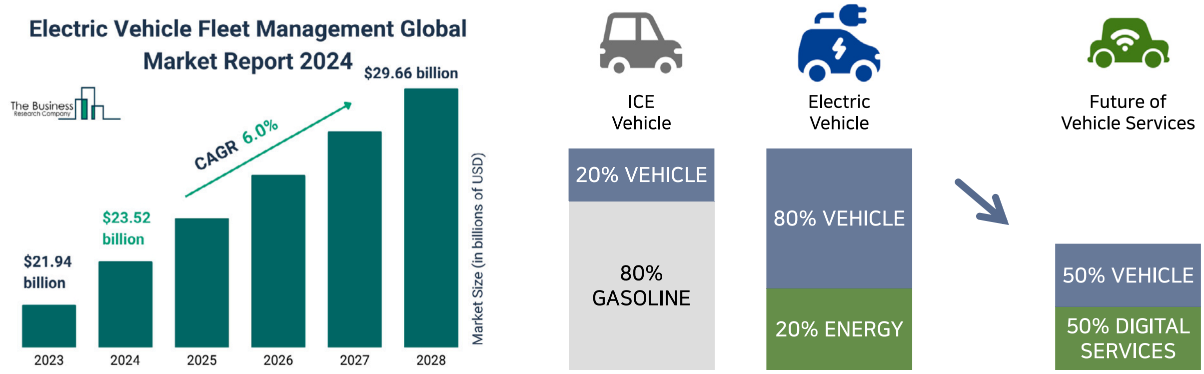


(a) (b)

Fig. 1. Growth of the EV market; (a) EV fleet management market size 2024 and growth rate [1], and (b) different ratios of capital expenditures (CAPEX) and operating expenses (OPEX) of traditional vehicles versus EVs [2].

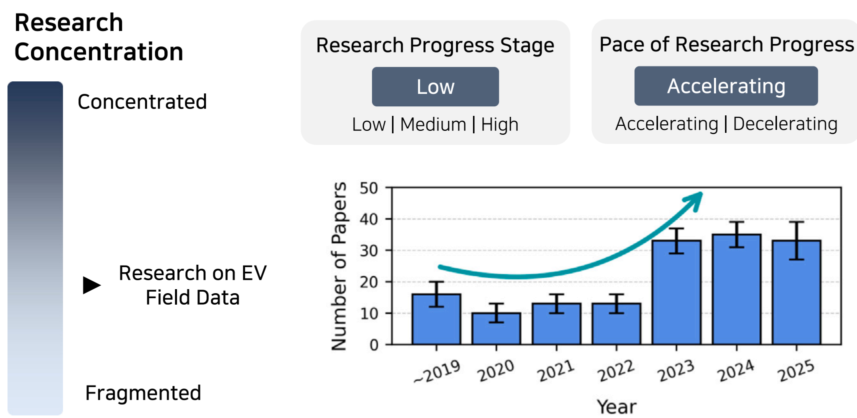


Fig. 2. EV field data research trends (keywords: electric vehicles, field data).

on studies utilizing real EV operational data reveals that publications surged from 13 in 2022 to over 30 per year since 2023, as illustrated in Fig. 2. This marks the transition of EV field data research from a nascent, fragmented area to a concentrated field with significant momentum. Pioneering research projects like FeBAL [13], BatteReverse [14], and OPEVA [15] exemplify this trend, focusing on building comprehensive analytics and lifecycle management frameworks.

Despite this momentum, the rapid expansion of field data research has resulted in a fragmented and challenging landscape. Significant hurdles related to data quality, standardization, and privacy continue to hinder progress and limit the validation of emerging methodologies. Consequently, a clear and structured overview is urgently needed to unify these disparate efforts and effectively guide future innovation. To the best of our knowledge, this paper presents the first systematic review dedicated exclusively to real-world EV field data, synthesizing essential insights by examining its transformative role. Specifically, this review surveys the current landscape of available datasets (Section 2), addresses critical challenges involved in transforming raw data into actionable insights (Section 3), explores key research applications (Section 4), and identifies promising future directions (Section 5). The overall structure of this review is summarized in Fig. 3. By integrating current progress and highlighting critical knowledge gaps, this paper aims to provide a comprehensive roadmap for leveraging field data, advancing battery technology, and accelerating the global transition toward sustainable mobility.

2. The landscape of EV field datasets

The effective use of EV field data begins with understanding its landscape, which spans from massive, industrial-scale data streams to curated, publicly available research datasets. This section categorizes these data sources, providing a foundational guide for researchers.

Standardized protocols are the backbone of large-scale data generation. China's GB/T 32960 protocol [16], for example, mandates comprehensive real-time monitoring, enabling the National Monitoring and Management Center for EVs (NMMCNEV) to amass approximately 20 TB of operational data daily—a scale reaching 7 PB by 2019 across over 6,700 vehicle types [17]. This massive data collection effort is supported by an extensive network of monitoring centers across China, including the Service and Management Center for electric vehicles (SMC-EV) [18–27], Beijing Electric Vehicles Monitoring and Service Center (BEVMSC) [28,29], the Shanghai Electric Vehicle Public Data Collecting, Monitoring and Research Center (SHEVDC) [30–34], the National Big Data Alliance for New Energy Vehicles (NDANEV) [35–45], the National Monitoring and Management Platform for Electric Vehicles (NMMP-EV) [46–48], and the National New Energy Vehicle Networking Platform (NNEVNP) [49]. To put this scale in perspective, this industrial dataset is roughly one million times larger than the well-known 7.8 GB MIT-Stanford battery aging dataset [50].

Beyond China, where such comprehensive operational data collection is mandated, global data protocols have primarily focused on standardizing the charging ecosystem. Standards such as ISO 15118 and OCPP 2.0.1 define the secure communication interface between EVs and charging stations, enabling features like Plug & Charge and smart grid integration. In North America, SAE J1939 provides a framework for data exchange on medium- and heavy-duty vehicles. While these protocols are crucial for charging analytics and interoperability, they do not currently facilitate the same scale of continuous, nationwide operational data logging seen under the GB/T framework.

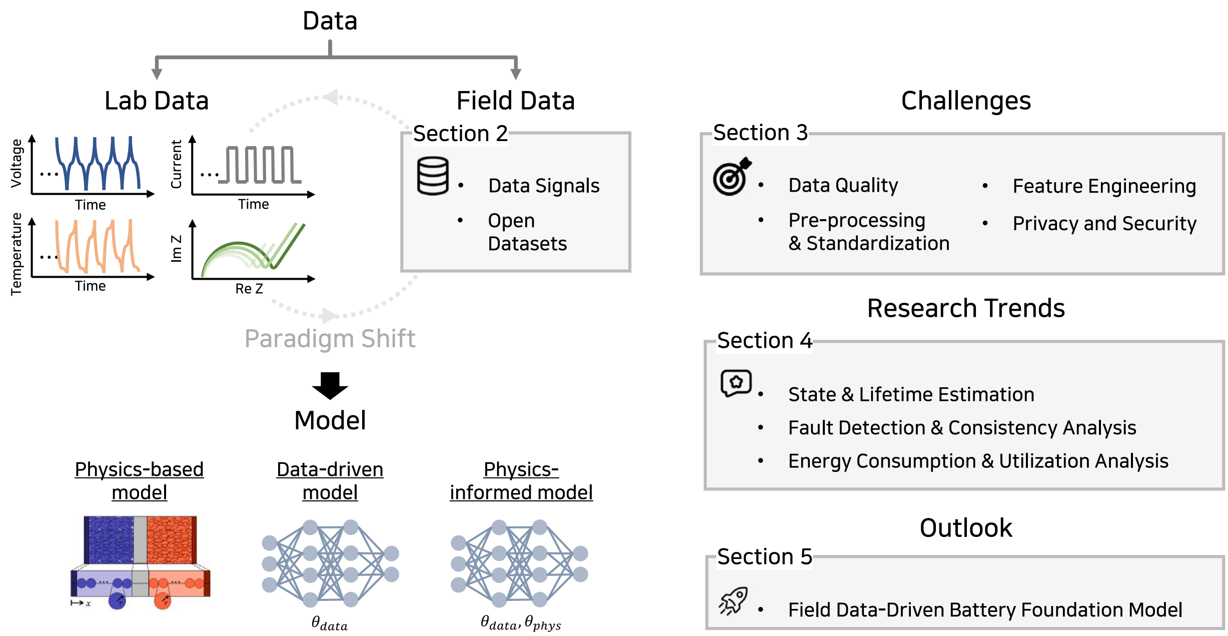


Fig. 3. Overall structure of this review paper.

Table 1
General vehicle and driving data fields.

Item	Description	Unit	Range/Format
`Vehicle_ID`	Unique vehicle identifier	–	–
`Time`	Timestamp of the data	–	yyyy/mm/dd hh:mm:ss
`Vehicle State`	Operational state	–	0: Drive, 1: Charge
`Event`	System event status	–	1: Normal, 2: Error
`Speed/emobility_spd`	Vehicle speed	km/h	0.0–120.0
`Accelerator Pedal Stroke`	Accelerator pedal position	%	0.0–100.0
`Brake Pedal Stroke`	Brake pedal position	%	0.0–100.0
`drive_motor_spd`	Speed of the drive motor	rpm	0–65535
`ext_temp`	Ambient (external) temperature	°C	−127–127
`int_temp`	Cabin (internal) temperature	°C	−127–127
`hvac_list`	HVAC (Heating, Ventilation, Air Con) data	°C	−127–127

Table 2
Core battery state and power data fields.

Item	Description	Unit	Range
`soc`	State of Charge	%	0.0–100.0
`soh`	State of Health	%	0.0–100.0
`socd`	Displayed State of Charge	%	0.0–100.0
`pack_current`	Current of the battery pack	A	−500.0–500.0
`pack_volt`	Voltage of the battery pack	V	0.0–1000.0
`batt_pw`	Power of the battery pack	kW	0–6553.5
`acceptable_chrg_pw`	Maximum acceptable charging power	kW	0–6553.5
`acceptable_dischrg_pw`	Maximum acceptable discharging power	kW	0–6553.5
`inverter_capacity_volt`	Voltage of the inverter capacitor	V	0–65535
`sub_batt_volt`	Voltage of the auxiliary battery	V	0–25.5

To interpret any of these datasets, a foundational understanding of the signals monitored by a vehicle's Battery Management System (BMS) is essential. While proprietary, these signals can be classified into key categories. Tables 1 through 6 provide a structured lexicon of these critical data fields, detailing core battery states (Table 2), cell-level granularity (Table 3), driving dynamics (Table 1), cumulative statistics (Table 4), system status flags (Table 5), and thermal management data (Table 6). This framework serves as a guide for navigating the public datasets now available to the research community.

Table 7 summarizes key public datasets that have been instrumental in recent research, with visualizations from select datasets shown in Figs. 4 and 5. Each dataset offers unique characteristics tailored to specific research questions:

Urban EV Data [45], Fig. 4(a): Distinguished by its immense scale, this dataset contains real-time operational data from over three million EVs in China. It provides an unparalleled resource for macroscopic analyses of urban mobility, fleet behavior, and large-scale infrastructure planning.

Multi-Modal Data [51], Fig. 4(b): Valuable for its long-term (3-year) perspective across 300 vehicles, this dataset is one of the few public sources containing granular cell-level voltage data for all 96 cells in the pack. This makes it ideal for developing robust SOH estimation models that account for cell-to-cell variations under realistic conditions.

Beihang Data [52], Fig. 5(a): This large-scale dataset from 514 vehicles also provides valuable cell-level data but presents a unique challenge: the variables are unlabeled and normalized. This makes it

Table 3
Battery cell and module data fields.

Item	Description	Unit	Range/Format
`max_cell_volt`	Maximum voltage	V	0.0–5.1
`min_cell_volt`	Minimum voltage	V	0.0–5.1
`max_cell_volt_no`	Cell number with the maximum voltage	Index	–
`min_cell_volt_no`	Cell number with the minimum voltage	Index	–
`cell_volt_list`	List of all individual cell voltages	V	Array of floats
`mod_max_temp`	Maximum temperature	°C	−127–127
`mod_min_temp`	Minimum temperature	°C	−127–127
`mod_avg_temp`	Average temperature	°C	−127–127
`mod_temp_list`	List of all individual module temperatures	°C	Array of integers

Table 4
Cumulative and lifetime data fields.

Item	Description	Unit	Range
`odometer/Mileage`	Accumulated total driving distance	km	0–16,777,215
`op_time`	Cumulative vehicle operation time	s	0–4,294,967,295
`cumul_current_chrgd`	Cumulative charged current	Ah	0–429,496,729.5
`cumul_current_dischrgd`	Cumulative discharged current	Ah	0–429,496,729.5
`cumul_pw_chrgd`	Cumulative energy charged	kWh	0–429,496,729.5
`cumul_pw_dischrgd`	Cumulative energy discharged	kWh	0–429,496,729.5
`trip_chrg_pw`	Energy charged during the current trip	kWh	0–6553.5
`trip_dischrg_pw`	Energy discharged during the current trip	kWh	0–6553.5
`chrg_cnt`	Total number of charging cycles	Count	0–4,294,967,295
`chrg_cnt_q`	Total number of fast charging cycles	Count	0–4,294,967,295

Table 5
System status and control flags.

Item	Description	Unit	Range/Format
`bms_running`	BMS operation status	–	0: Off, 1: On
`main_relay_conn`	Main relay connection status	–	0: Disconnected, 1: Connected
`slow_chrg_port_conn`	Slow (AC) charge port connection status	–	0: Disconnected, 1: Connected
`fast_chrg_port_conn`	Fast (DC) charge port connection status	–	0: Disconnected, 1: Connected
`fast_chrg_relay_on`	Fast charge relay status	–	0: Off, 1: On
`chrg_cable_conn`	Charging cable connection status	–	0: Disconnected, 1: Connected
`batt_fan_running`	Battery cooling fan status	–	0: Off, 1: On
`insul_resistance`	Insulation resistance of the high-voltage circuit	MΩ	0–65.535
`est_chrg_time`	Estimated remaining time for full charge	s	0–4,294,967,295

Table 6
Thermal management system data fields.

Item	Description	Unit	Range
`batt_internal_temp`	Internal temperature of the battery pack	°C	−127–127
`batt_coolant_inlet_temp`	Temperature of the coolant at the battery inlet	°C	−127–127
`batt_pra_busbar_temp`	Temperature of the PRA (Power Relay Assembly) busbar	°C	−127–127
`batt_ltr_rear_temp`	Temperature at the rear of the LTR (Low Temp Radiator)	°C	−127–127

particularly suitable for developing and testing advanced, unsupervised feature extraction and diagnostic techniques.

Volkswagen (Audi e-tron) Data [53], Fig. 5(b): In contrast to fleet-level data, this dataset provides an unparalleled, high-frequency (100 Hz) view into a single Audi e-tron over one year. Its 2 TB of raw, unprocessed signals offer a unique opportunity for detailed analysis of transient battery dynamics during real-world driving and charging.

BAIC Charging Data [54], Fig. 6(a): Collected from 20 commercial EVs over 29 months, this dataset's primary focus is on charging data. It captures real-world multi-stage constant current (MCC) charging patterns, making it highly relevant for studying the long-term effects of different charging strategies on battery health.

Tsinghua Fault Data [55], Fig. 6(b): This dataset is uniquely designed for diagnostics, offering 690,000 charging snippets from 347 EVs with verified, ground-truth fault labels. Due to the normalization of all signals for privacy, it is an essential resource for training and validating advanced, data-driven fault detection models that go beyond simple thresholding.

Commercial EV Data [56], Fig. 7(a): Focusing on specific commercial use cases, this dataset provides a direct comparison between EVs used in geriatric care and carsharing fleets versus a private commuter vehicle. A notable feature is that data was logged only during driving. This is complemented by a separate, supplementary database of complete charging curves recorded under controlled conditions. This unique structure makes the dataset ideal for analyzing application-specific usage profiles, while also providing distinct data for studying charging characteristics.

e-Transportation Data [57], Fig. 7(b): This dataset provides rare insights into the operational profiles of non-car EVs, specifically 82 electric buses and six electric boats. It is a key resource for understanding the unique battery stress factors and energy consumption patterns in heavy-duty and marine transport applications.

BMW i3 Data [58], Fig. 8(a): This dataset from a single BMW i3 is specifically tailored to analyze the impact of seasonal conditions and HVAC usage on battery performance. Its strength lies in the inclusion of extensive thermal management signals, making it ideal for modeling the effects of auxiliary power consumption on vehicle efficiency.

GZ NEIV Data [59], Fig. 8(b): This dataset presents operational data from 10 diverse EVs, including 7 passenger cars (typically with 91 cells, 150–160 Ah capacity) and 3 electric buses (with 324–360 cells, 50–645 Ah capacity). Acquired via T-BOX, it includes key variables like speed, mileage, SOC, current, voltage, and min/max cell temperatures.

Table 7
Overview of open field datasets.

Dataset	Spec.	Data scope	Sampling	Format	Norm.	Remarks
Urban EV [45] URL	Various EVs, NMC Pouch	3 million+ vehicles (2018–2019)	1–30 s, 1–10 m	.csv	Raw	Large-scale urban data for macroscopic analysis
Multi-Modal [51] URL	300 EVs, 155 Ah NCM	Over 3 years	0.1 Hz	.csv	Raw	Long-term dataset with cell-level voltage
Beihang [52] URL	514 EVs (3 OEMs)	18.2 million entries	0.033 Hz	.pkl	Normalized	Cell-level data; unlabeled variables
Volkswagen e-tron [53] URL	95 kWh, Pouch (432 cells)	1 vehicle over 12 months (2019–2020)	100 Hz	.mat	Raw	High-frequency e-SUV data
BAIC Charging [54] URL	20 BAIC EU300, 45.3 kWh NMC	29 months of charging data	0.125 Hz	.csv	Raw	Battery aging during charging
Tsinghua Fault [55] URL	347 EVs (3 brands)	Up to 30 months of charging data	–	.pkl	Normalized	Labeled real-world charging faults
Commercial EV [56] URL	9 EVs (Smart e.d., iMiEV, iOn), NMC	14–22 months (2014–2015)	1 Hz	.parquet	Raw	Application-specific use-case analysis
e-Transportation [57] URL	82 e-Buses (NMC, LMP), 6 e-Boats	Up to 14 months	0.1, 0.2 Hz	.csv	Raw	Heavy-duty, marine transport
BMW i3 [58] URL	33 kWh, 60 Ah, Prismatic NMC	1 vehicle, 70 trips (2019–2020)	10 Hz	.csv	Raw	Seasonal driving and HVAC data
GZ NEIV [59] URL	10 EVs (7 passenger, 3 bus), NCM, LFP	1 month each	0.1, 0.5 Hz	.xlsx	Raw	(full year available on request)

Table 8
The cost of battery SOH evaluation for EVs [51].

Testing expense (per vehicle)					
Item	Man-hour	Quantity	Unit	Unit price (¥)	Total price (¥)
SOH evaluation	9	1	/Hour	200	1800
Technical support	/	1	/Vehicle	500	500
Venues expense	/	1	/Day	200	200
Sum up			/ Vehicle		2500
Manpower/equipment expense (per day)					
Manpower and management	/	1	/Person*day	300	/
Equipment adjustment and depreciation	/	1	/Set*day	400	/

The publicly available portion covers approximately 12.8 million data points across one month per vehicle, with cumulative mileages ranging from roughly 27,000 to 132,000 km. A full year's data per vehicle can be provided upon request, making it a valuable resource for analyzing battery performance and thermal management across various EV types.

3. From raw data to actionable insights: Challenges and methodologies

Leveraging EV field data requires navigating a series of significant challenges that are fundamentally different from those in controlled laboratory settings. As illustrated in Fig. 9, these hurdles span the entire data lifecycle, from ensuring data quality and pre-processing raw signals to engineering meaningful features and managing privacy. Overcoming these obstacles is critical to unlocking the full potential of real-world battery analytics.

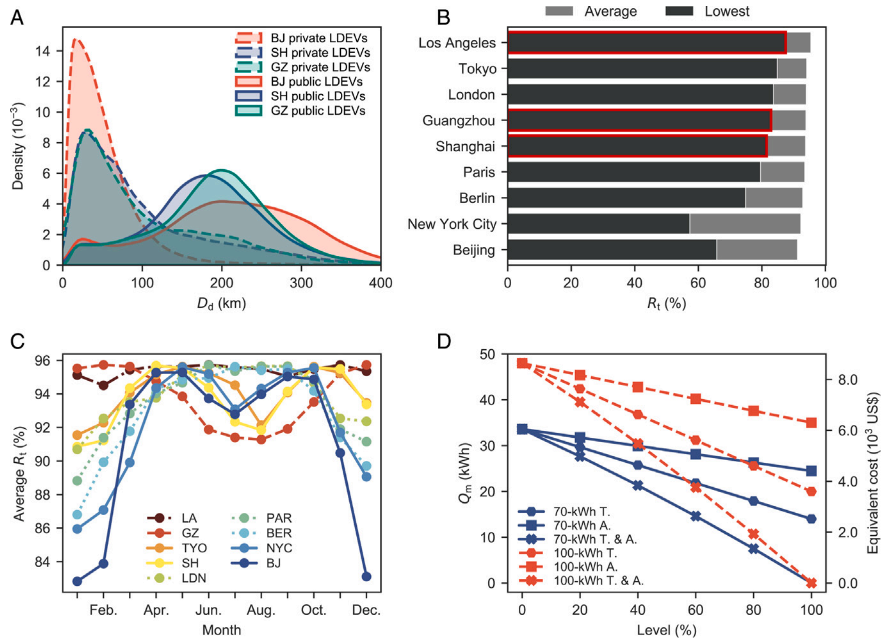
3.1. The data quality challenge

The foremost challenge lies in data quality and the ambiguity of ground truth when defining reliable State-of-X (SOX) metrics. Here, SOX is a general term that includes not only commonly used SOC and SOH but also other indicators such as state of power (SOP), state of energy (SOE), state of function (SOF), and state of safety (SOS). Unlike in a lab where SOH is measured with full charge–discharge cycles [51], a procedure that is fundamentally incompatible with real-world EV usage patterns [60–65]. Researchers are left with the onboard SOC as a proxy, but it is often an unreliable and drifting metric, creating a recursive uncertainty problem where no stable baseline for degradation analysis exists [66].

This issue is compounded by the corruption of classic diagnostic signatures; the clean Incremental Capacity (IC) or Differential Voltage (DV) peaks seen in lab data become distorted and unrecognizable in the noisy, dynamic thermal and electrical conditions of real-world operation [67]. Furthermore, pack-level data is governed by the "tyranny of the weakest cell" [68–70], where manufacturing variations and uneven aging cause a single underperforming cell to dictate the operational limits of the entire pack [35,52], making a unified "pack SOH" a dangerously misleading abstraction.

At the same time, the irregular pulses and dynamic load variations inherent in real driving can serve as natural excitation signals. These signals may open up new possibilities for alleviating identifiability issues in physics-based models [71–74], providing parameter pathways that are difficult to access under conventional, static laboratory protocols. In this sense, field datasets – despite their noise and incompleteness – hold the potential to yield distinctive system-level insights that are unattainable from laboratory data alone.

Nevertheless, significant limitations remain, underscoring not only the challenges of interpreting field data but also the scarcity of reliable datasets. An emerging approach to address data scarcity is the generation of synthetic data. By creating artificial degradation trajectories and operating scenarios, such datasets provide an intermediate layer between highly controlled laboratory measurements and sparse, noisy field signals. Early pioneering work [75,76] and later extensions to diverse chemistries and even battery packs [77–81] have shown the value of synthetic data for testing diagnostic and prognostic tools under



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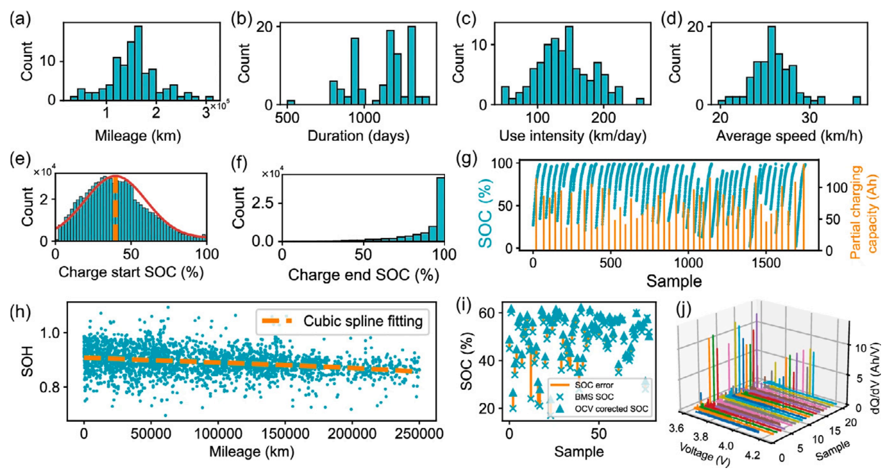


(b)

Fig. 4. Data visualization; (a) Urban EV data [45]; **A** daily driving distance distributions, **B** annual average and lowest upper limits of battery utilization rates (Rt), **C** monthly average upper limits of battery utilization rates, **D** maximum unavailable battery energy (Qm) and equivalent battery costs in Beijing with varied battery energy (70 kWh and 100 kWh) under individual or combined (T & A), and (b) Multi-Modal data [51]; **a** distribution of mileage, **b** distribution of duration, **c** distribution of use intensity quantified by driving distance per day, **d** distribution of the average driving speed, **e** distribution of charging start SOC, **f** distribution of charging end SOC, **g** EV charging example, **h** SOH degradation, **i** SOC estimation error of BMS, **j** IC analysis for EV charging process.

varied degradation scenarios at low cost. Synthetic data thus serves as a practical bridge across the lab–field spectrum. In the long run, however, as large-scale field datasets accumulate and naturally capture a wide range of conditions (e.g., C-rates, temperatures, usage cycles), this gap will be progressively closed and the reliance on synthetic data will diminish.

3.2. Pre-processing and standardization

A foundational challenge in utilizing field data is the pre-processing required to create a clean, analyzable dataset from raw, chaotic logs [37,82–84]. As illustrated in the pipeline in Fig. 10, this requires a pipeline that ingests thousands of files (e.g., MDF, CSV) and performs operational segmentation—partitioning the continuous data stream into distinct states like charging or driving. This process must immediately resolve timestamp misalignment from asynchronous ECU logging, a common issue during events like multi-stage charging protocols where current and voltage signals are not recorded simultaneously [6]. Sophisticated resampling and interpolation are necessary to create the coherent data segments that form the basis for any valid analysis [85,86].

The resulting high data volume then necessitates intelligent data reduction using specialized algorithms – such as Largest-Triangle-Three-Bucket (LTTB) [87] for preserving transient peaks or Savitzky–Golay filters for trend analysis – to avoid destroying diagnostically significant features [48,88–90]. The choice of downsampling method is a critical trade-off between reducing data volume and preserving diagnostically significant phenomena.

Beyond cleaning individual datasets, the field data is hampered by a systemic standardization crisis [91], a lack of common data formats, sampling rates, and performance benchmarks makes objective comparison across studies nearly impossible and necessitates the creation

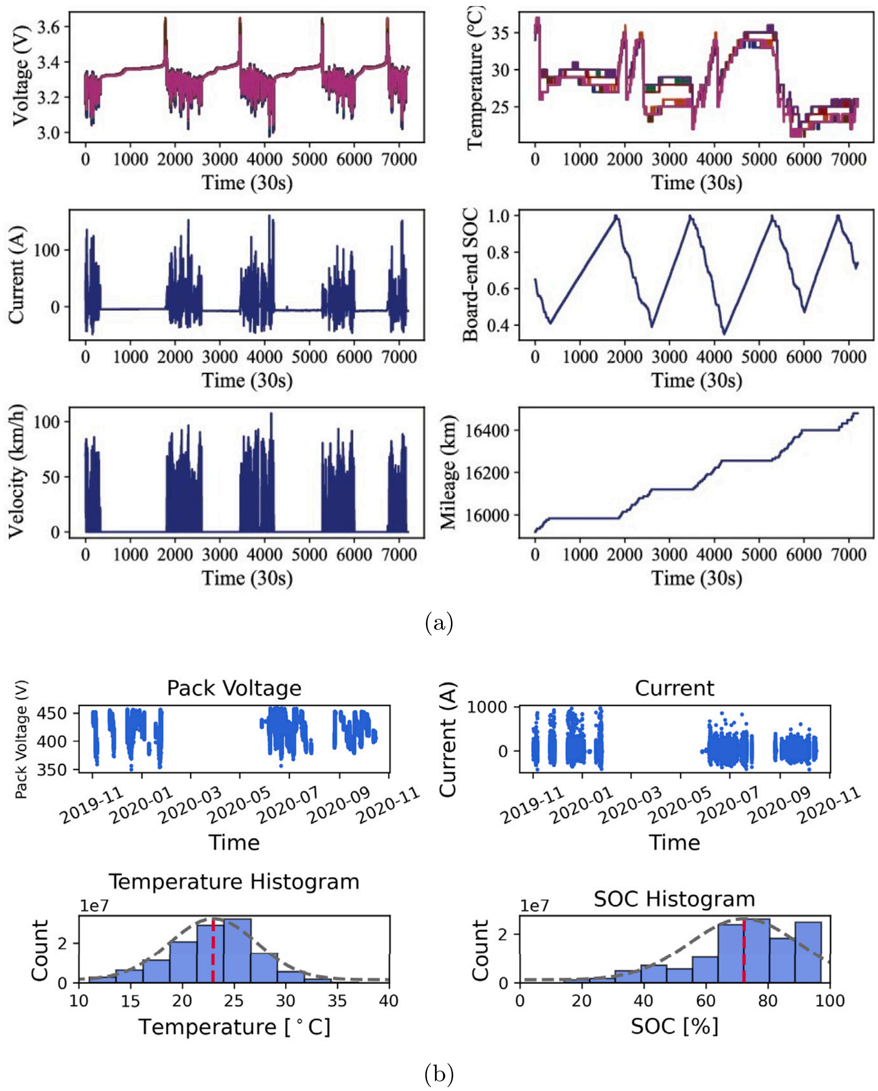


Fig. 5. Data visualization; (a) Beihang data [52]; sample sequences of voltage, temperature, current, speed and mileage, and (b) Volkswagen e-tron data; pack voltage and current values, temperature and SOC distribution across the entire interval.

of harmonized databases [56,84,92–94]. Finally, pre-processing must account for structural imperfections designed into the BMS, including systematic data gaps when the vehicle is off and sparse data logging. Due to cost and data storage constraints, the BMS often records only the maximum and minimum cell voltages or temperatures – a consequence of sparse sensor placement with sensor-to-cell ratios as low as 1/10 as shown in Fig. 11 – rather than the full data distribution, fundamentally limiting the granularity of any analysis.

3.3. Feature engineering

Classic battery feature engineering, particularly methods like IC and DV analysis, is fundamentally designed for laboratory settings. These techniques depend on clean, monotonic voltage curves generated during slow, constant-current charging to extract stable physical signatures [67]. Such controlled conditions, however, are rarely met in the field, causing these lab-centric methods to fail. Real-world data is dominated by multi-stage fast charging, which introduces non-monotonic voltage profiles heavily distorted by battery polarization and dynamic current changes [96,97]. This operational variability, compounded by diverse user behaviors (e.g., charging depth, rates) and ambient conditions, renders the delicate physical signatures unstable and unreliable, necessitating a shift toward feature engineering specifically tailored for the challenges of field data [98].

A common pragmatic response is to rely on robust statistical features [5,84,99–103], though this often risks oversimplifying complex degradation processes. Advanced feature engineering is therefore branching out in several directions to tackle the unique challenges of field data. The first strategy involves defining new, physically meaningful performance indicators from collected operational data. For example, Pozzato et al. [53] compute driving resistance from brief acceleration/braking peaks and a novel charging impedance that serves as a system-level proxy for DV analysis (Fig. 12(a)). However, a key challenge is that these indicators are highly sensitive to seasonal temperature, making it difficult to disentangle thermal effects from aging-related degradation. The second direction focuses on automated frameworks that synergistically combine lab and field data, especially when field data is only available as memory-saving histograms. Steininger et al. [104] propose a workflow that uses reliable correlations from lab data to guide feature selection from noisy field data, then uses functional fitting to reconstruct richer information from the coarse histogram bins (Fig. 12(b)). The third frontier moves from passive observation to active diagnostics [105]. This method involves applying a specific DC electrical stimulus to the battery to intentionally trigger a time-varying response. From this response, a hierarchy of

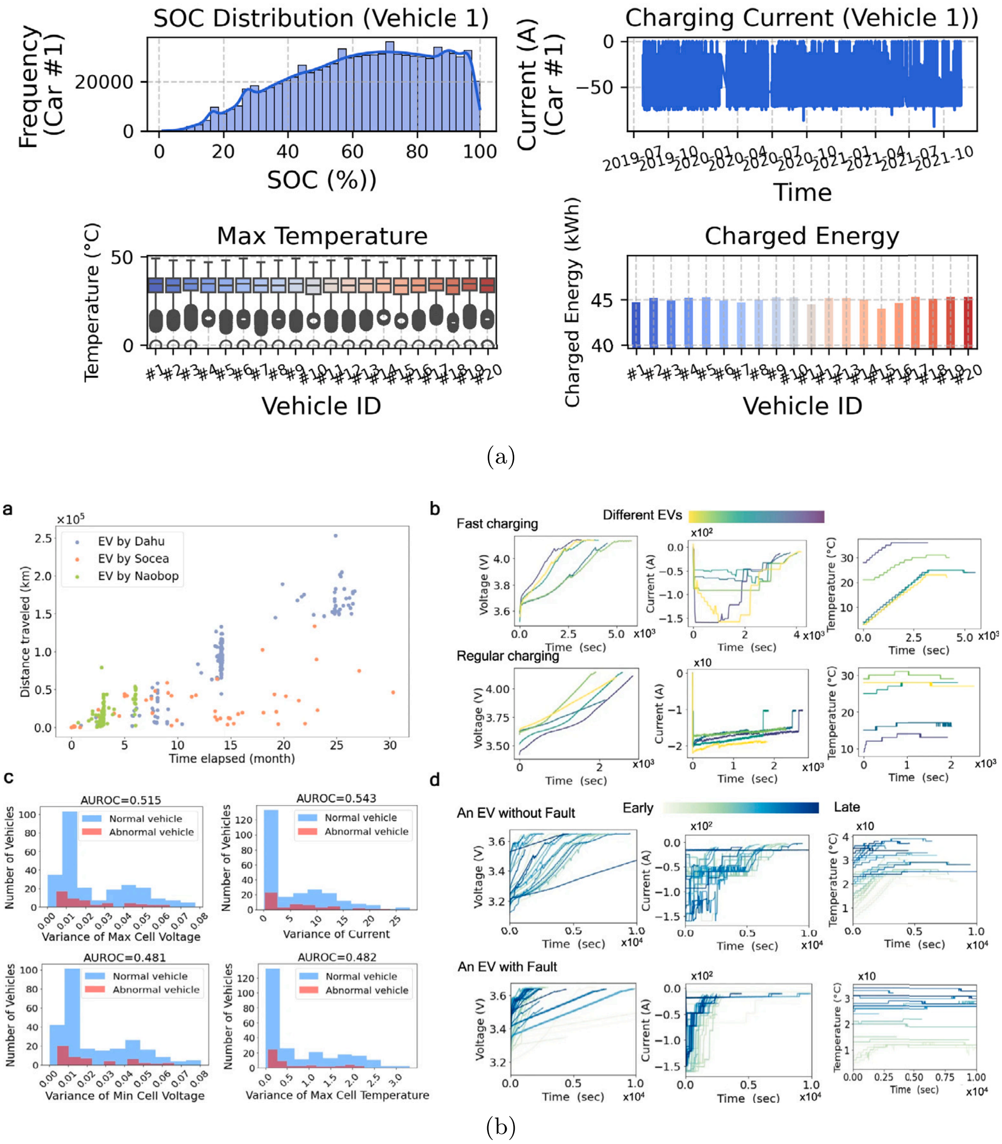


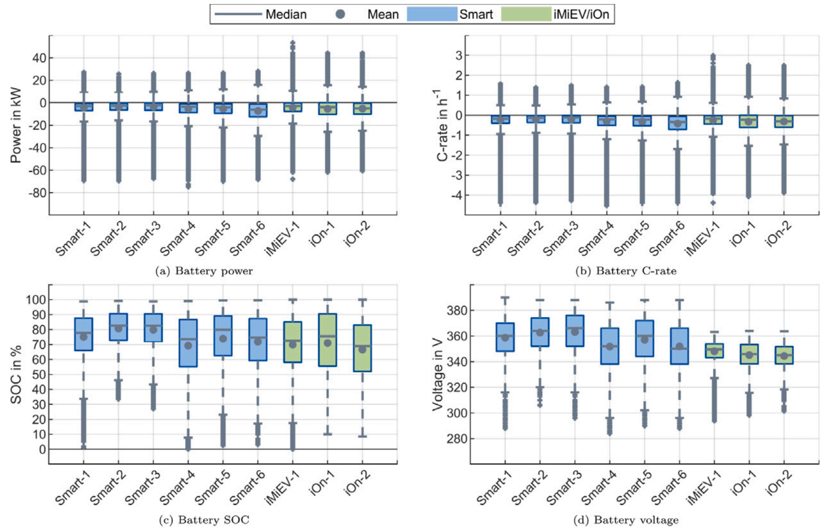
Fig. 6. Data visualization; (a) BAIC Charging data; SOC distribution and charging current for car #1, maximum temperature and charging energy for all vehicles, and (b) Tsinghua Fault data [55]; **a** distance traveled over the time elapsed of three manufacturers, **b** sampled charging segments of fast charging and regular charging, **c** normal and abnormal EVs AUROC, **d** a detailed view of charging snippets.

primary, secondary, and ultimately composite response parameters are derived, which are engineered to be exceptionally sensitive to the early-stage precursors of internal faults. Ultimately, feature engineering must evolve beyond focusing on individual sensor values and towards modeling the physical coupling and spatial topology between sensors, thereby capturing the relational integrity of the entire system (Fig. 12(c)).

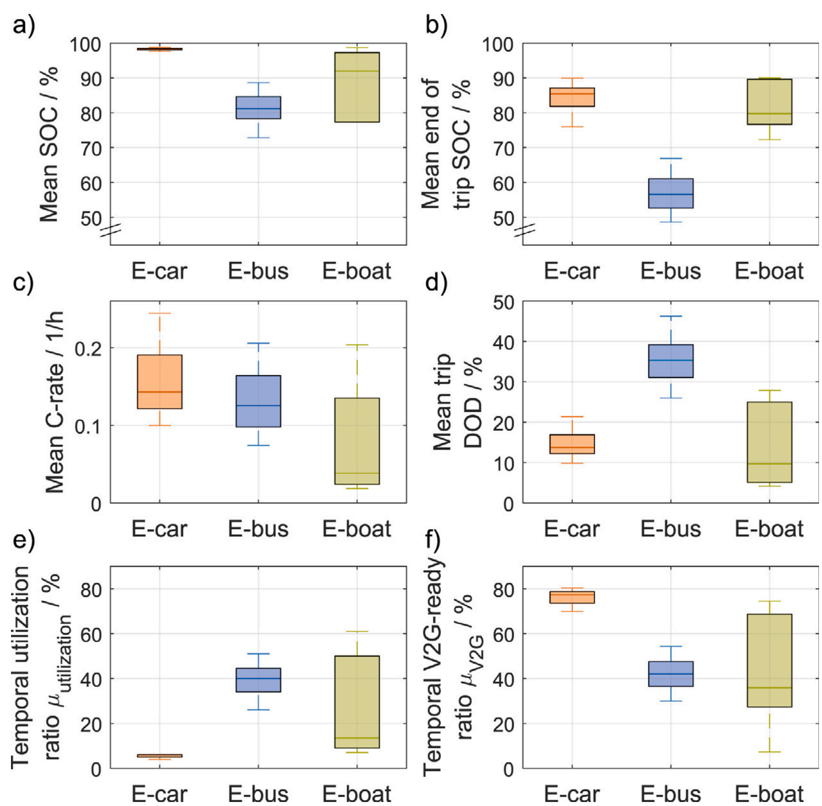
3.4. Privacy and security

The core challenge in battery field data is a dual-fronted conflict between diagnostic utility and data sharing. On one hand, the most valuable data for prognostics – high-frequency current, voltage, and location-correlated events – creates a detailed digital footprint of a driver's behavior [55,63,106]. On the other, the reluctance of OEMs to share valuable, proprietary battery data with third-party services presents a major commercial barrier to collaborative research [94]. Consequently, even when datasets are made public, they are often processed to protect privacy and intellectual property, for instance, by removing explicit variable names or normalizing the data. This dual challenge of user privacy and corporate sensitivity necessitates advanced solutions that go far beyond simple anonymization.

The most powerful emerging solution is not a single algorithm but an edge–cloud collaborative architecture. This framework intelligently divides labor: the on-board (edge) system in the vehicle handles immediate, high-frequency processing for real-time fault detection, while the cloud performs computationally expensive, long-term analysis [107]. A key innovation is the encoder–decoder deployment: an encoder on the edge compresses raw data into a low-dimensional, less-sensitive representation before transmission [55]. This architecture provides a threefold benefit: it minimizes data transfer, protects the proprietary model on the cloud, and shields the user's raw, sensitive data from exposure. This architecture is enabled by innovations in both algorithms and protocols. For model training, Federated Learning (FL) is a key technology. By training models directly on each vehicle's BMS and only sharing aggregated model updates, FL allows for collaborative learning



(a)



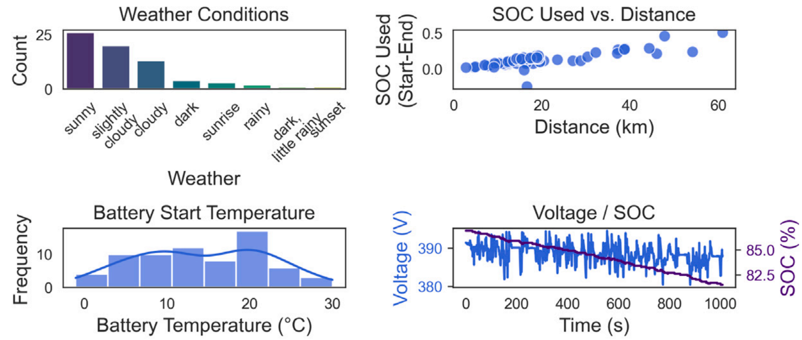
(b)

Fig. 7. Data visualization; (a) Commercial EV data [56]; boxplots of recorded values (power, C-rate, SOC, voltage) of all vehicles, and (b) e-Transportation data [57]; e-Car (60, simulated), e-Bus (52, field data), and e-Boat (6, field data) boxplots.

without centralizing raw data, permitting local models to adapt to specific conditions while benefiting from fleet-wide knowledge [108]. Beyond model training, ensuring the integrity and controlled sharing of data itself is critical. Here, Blockchain technology offers a complementary solution [94]. Its decentralized ledger and smart contracts can create a secure and transparent environment for data exchange between diverse stakeholders, enforcing access rights and creating a trusted ecosystem for collaborative research.

4. Key research trends

Research leveraging real-world field data has grown exponentially, a trend clearly illustrated by the increasing volume of publications in recent years (Fig. 13(a–b)). A critical factor within this research is data resolution, as the sampling time of field data is often limited by the data source (e.g., the vehicle's manufacturer) and can even vary for different signals within the same dataset. An analysis of the sampling



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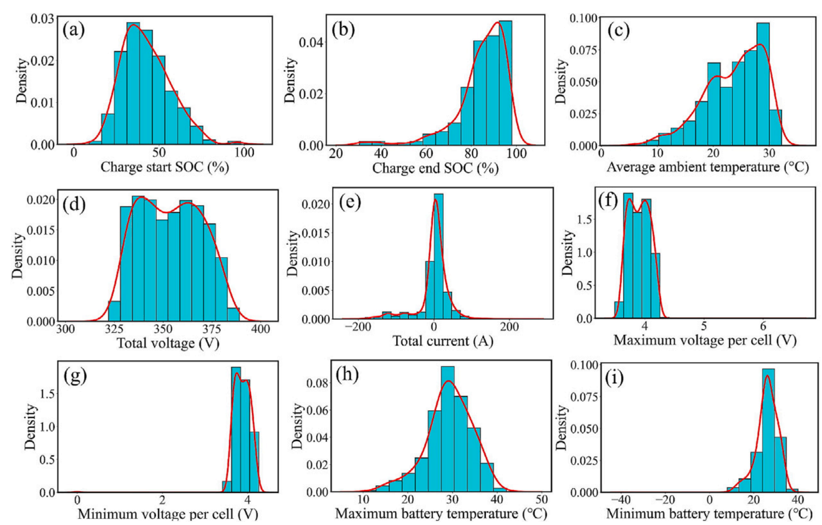


(b)

Fig. 8. Data visualization; (a) BMW i3 data [58]; frequency of weather conditions, SOC used vs. distance, distribution of battery start temperatures of TripA01 data, voltage and SOC of TripA01 data, and (b) GZ NEIV data [59]; raw data distributions for Vehicle #1.

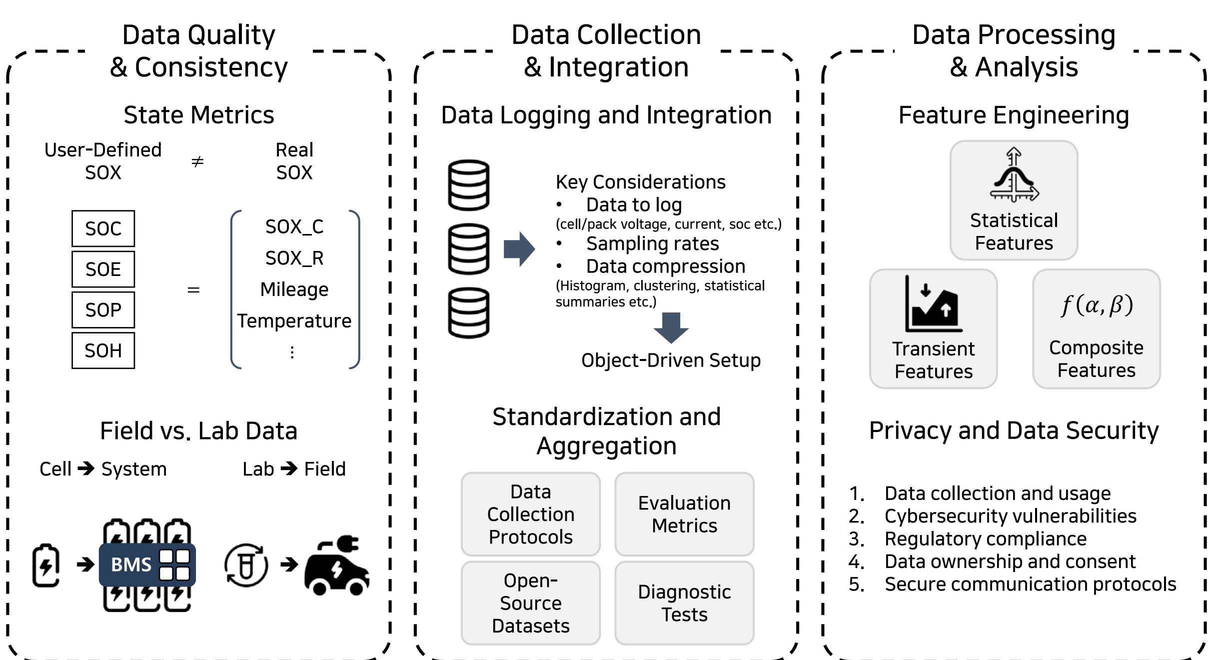


Fig. 9. Key challenges for field data.

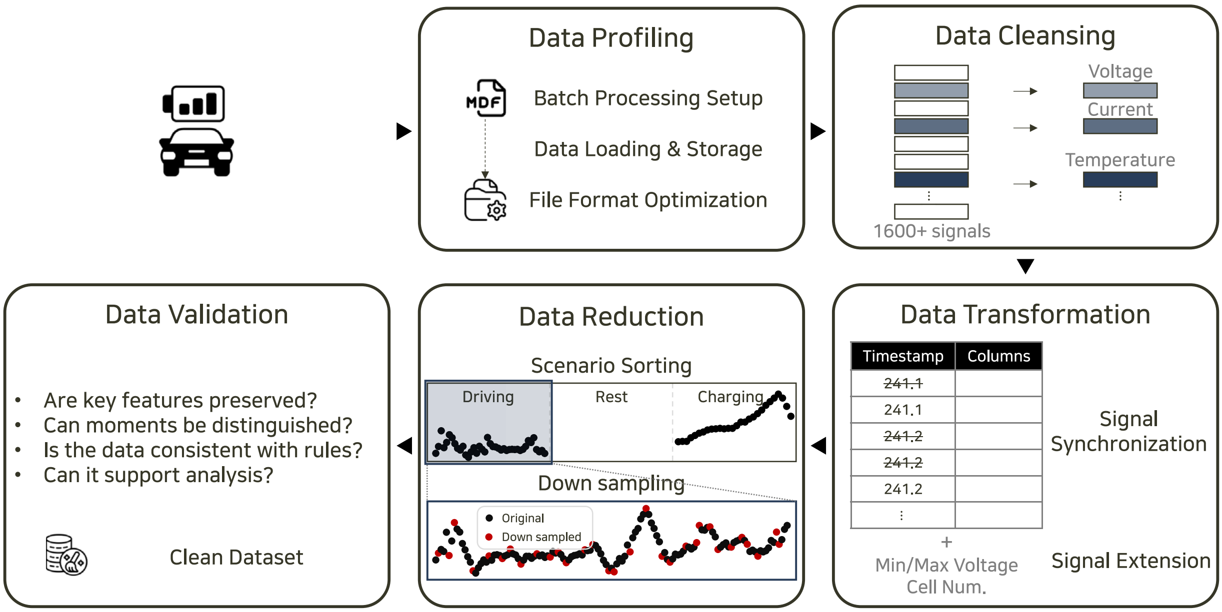


Fig. 10. Data pre-processing pipeline.

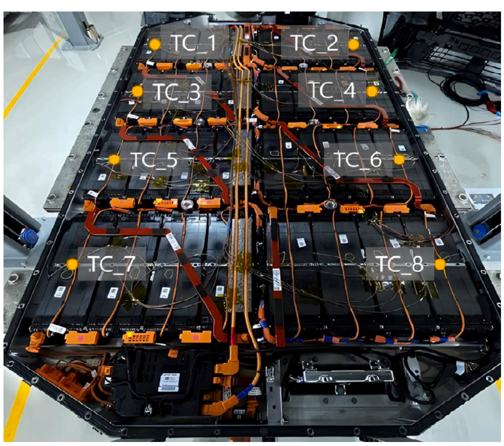


Fig. 11. Location of thermo-couple (T/C) sensors in the EV battery pack [95].

times used reveals distinct patterns across different applications (Fig. 13(c–d)). The following sections will explore how this field data is being applied across the primary research domains: Battery State and Lifetime Estimation, Fault Detection and Consistency Analysis, and Energy Consumption and Utilization Analysis.

4.1. Battery state and lifetime estimation

Field data has fundamentally reshaped battery state estimation, forcing a move from models based on idealized lab conditions to frameworks that can navigate the complexities of real-world operation. While this has spurred the development of purely data-driven methods, it has also highlighted the critical need to integrate these approaches with electrochemical models. This hybrid strategy leverages the statistical power of machine learning (ML) and the physical interpretability, leading to more robust and accurate state estimation across the primary tasks: SOH, SOC, Voltage/Temperature, and Remaining Useful Life (RUL) prediction. A systematic flowchart of this overall framework is provided in Fig. 14, which illustrates the main pipeline from data acquisition to model training and adaptive learning.

SOH estimation from field data is primarily a two-stage process due to the absence of ground-truth capacity labels. For EVs, SOH is most practically defined by capacity fade, as this directly impacts driving range [61,97,109,110]. However, performing a full laboratory-style evaluation is often unfeasible. Traditional workshop-based SOH evaluation, which involves a full discharge and a slow, methodical recharge, is not only time-consuming – often exceeding eight hours per vehicle – but also economically burdensome. The associated costs, which include manpower, equipment, and testing expenses, can be substantial, reaching an estimated ¥35,200 ($4,915) for a small batch of 10 vehicles and escalating to around ¥375,000 ($52,350) for 100 EVs (Table 8). These practical constraints make frequent, large-scale SOH benchmarking impractical.

Consequently, field-based research focuses on pseudo-ground truth labeling from noisy, incomplete operational data. Key methods include adapting Ampere-hour integration to stable charging segments [82,111,112] and applying Open Circuit Voltage (OCV) corrections [65,113,114]. A particularly powerful field-data technique is the use of statistical aggregation [54,60,62]; for instance, taking the monthly median of calculated capacity values creates robust labels that are resilient to the inherent noise and outliers in real-world data streams.

The second stage uses these generated labels to train predictive models. The richness of field data enables the extraction of diverse feature categories that are impossible to obtain in a lab, including: (i) cumulative parameters like total mileage and Ah-throughput [40,46,88,115,116]; (ii) operational stress factors such as temperature and depth-of-discharge (DOD) [51,117,118]; (iii) instantaneous characteristics extracted from charging processes, including peak/valley voltages, slopes of charging voltage curves, and real-time impedance measurements that provide immediate insights into battery state changes [5,36]; (iv) statistical features like the variance and kurtosis of voltage signals that capture operational variability [84,96,108,119,120]; and (v) cell inconsistency metrics that track voltage and temperature differences across the pack. These features are often combined with established Health Indicators (HIs) from IC Analysis (ICA) or DV Analysis (DVA), which correlate directly with internal degradation modes like Loss of Lithium Inventory (LLI) and Loss of Active Material (LAM) [121].

A parallel and crucial line of research focuses on full OCV reconstruction, which provides much deeper diagnostic capability than a single SOH percentage by enabling differentiation of key degradation modes. Significant work from the BMW Group and University of Bayreuth [79,98,122] demonstrates their ΔQ-method, a physics-based

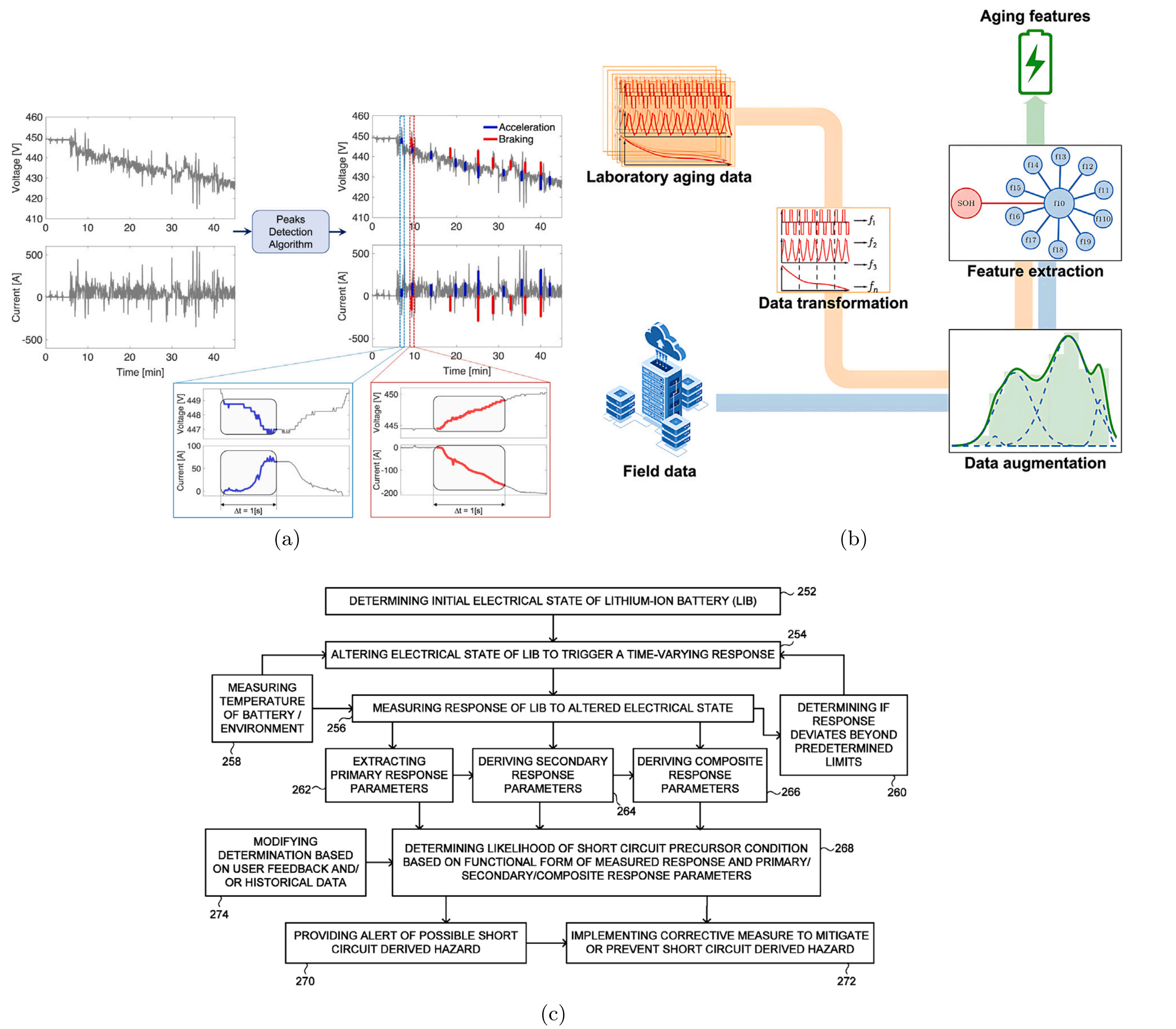


Fig. 12. Feature engineering strategies; (a) physically meaningful indicators using a peak-detection algorithm [53], (b) a framework combining laboratory and field data [104], and (c) composite response parameters [105].

approach that reconstructs full OCV curves using sparse field data from just a few relaxed voltage points in a BMW i3 fleet, achieving SOH errors below 3%. This work has evolved to integrate data-driven techniques, using transfer learning (TL) to adapt models with minimal data (Fig. 15(a)). The latest advance, physics-constrained transfer learning, combines a TCN-LSTM neural network with a mechanistic model, creating a hybrid approach that enhances stability and accuracy even with very limited fine-tuning data.

Beyond individual case studies, recent studies have utilized publicly available field datasets to develop and validate diverse SOH estimation models [54,60,124–128]. Table 9 provides a comparative summary of these models, illustrating how open field data has enabled systematic benchmarking across methods, and allowing a clearer analysis of trade-offs in uncertainty quantification (UQ), dependence on laboratory calibration, and achieved performance.

More recently, diagnostic-free approaches have emerged, where interpretable machine learning models can directly infer health states and electrode-level aging modes from random portions of operational data without requiring dedicated diagnostic cycles [123], highlighting a promising direction for practical onboard health assessment (Fig. 15(b)). In addition, current advances in self-supervised learning, domain adaptation, and continual learning are showing strong potential for improving generalization to unlabeled, heterogeneous, and evolving field datasets, thereby complementing semi-supervised and transfer learning strategies in real-world scenarios. By leveraging these advances, overfitting to specific fleets and susceptibility to dataset shift can be alleviated, enabling models that remain reliable across diverse operating conditions [129–132].

While SOH and OCV reconstruction represent the frontier of detailed diagnostics, the estimation of SOC and Voltage/Temperature remains a fundamental challenge in the field. The dynamism and noise of real-world data make it difficult for traditional Kalman filters to maintain accuracy [24]. Modern deep learning models [25,116,133,134], particularly Long Short-Term Memory (LSTM) networks, Generative Adversarial Networks (GANs), and Transformer networks, are better suited as they can learn the complex, non-linear dynamics directly from vast fleet datasets without relying on fixed, lab-derived parameters.

Finally, predicting RUL from field data requires a paradigm shift from lab-based cycle life prediction. The focus moves to user-centric

Fig. 13. Research trend; (a) overall publication proportions (b) flow of publication trends in EV battery research (c) publication distribution by sampling time bins (d) proportion of sampling times used within each research category.

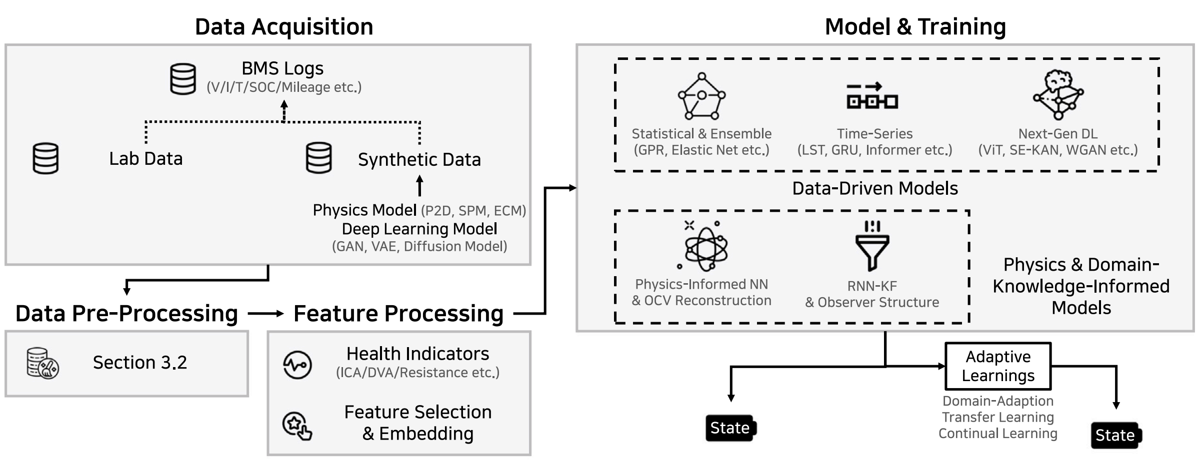
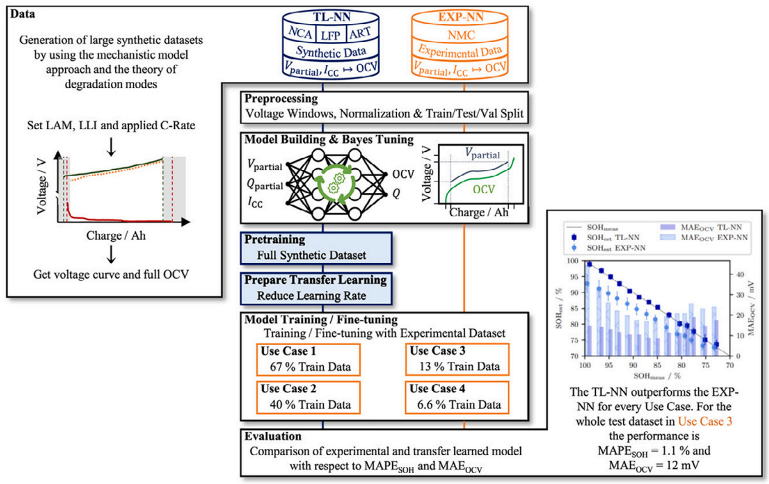


Fig. 14. Systematic flowchart of the battery state and lifetime estimation.

metrics like remaining kilometers or years of service. Most importantly, large-scale field data reveals that real-world degradation is both stochastic and non-linear, often proceeding in distinct phases: a study identified an initial rapid decline (e.g., 0–77,000 km), a stable mid-life period (e.g., 77,000–199,200 km), and a slower late-life stage [66,83]. Given this complexity and uncertainty, the research frontier is focused on probabilistic frameworks that can provide a reliable confidence interval for the RUL [135,136], which is far more valuable for real-world decision-making than a single deterministic prediction. The thematic evolution of these field-data-based estimation techniques – from adapting to real-world noise to enhancing model trustworthiness – is outlined in Table 10.

4.2. Fault detection and consistency analysis

Fault diagnosis in the field operates under a fundamentally different paradigm than in the laboratory. The extreme scarcity of labeled failure data makes traditional supervised classification impractical. Consequently, the focus has shifted to unsupervised anomaly detection [41,



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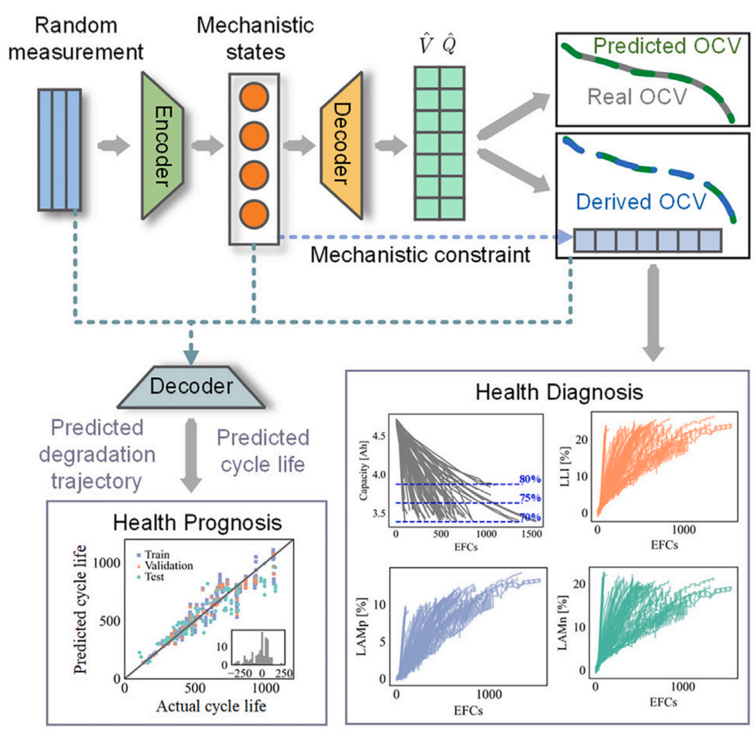


(b)

Fig. 15. Battery state and lifetime estimation research; (a) transfer learning from synthetic data for OCV reconstruction and SOH estimation [79], and (b) diagnostic-free onboard model for battery health assessment and degradation forecasting [123].

103,137–141]. The objective is to build a robust model of "normal" battery behavior and identify deviations. Crucially, field data reveals that catastrophic failures are not always triggered by acute events but can arise from the slow accumulation of minor stresses – such as gradual electrolyte degradation or slight overcharging – that progressively reduce the battery's stability margin over time [142–146]. The goal is therefore to detect this slow drift towards instability, whose signatures vary significantly by fault type: a developing thermal runaway may present as a gradual temperature rise and voltage instability, whereas an internal short circuit or electrolyte leakage often presents as more subtle but persistent voltage volatility and noise [4,22,52,147] as shown in Fig. 16.

To systematically organize these diverse approaches, fault detection and inconsistency analysis can be categorized into four methodological families: residual-driven, feature-driven, distribution-driven, and physics/knowledge-driven methods, as summarized in Fig. 17.

Beyond methodological taxonomy, it is also important to compare how models perform on the same publicly available field datasets.

Table 9
Comparison of battery SOH estimation models using real-world and laboratory data. Rows highlighted in gray denote studies that publicly released their datasets, which were subsequently utilized for literature-based comparative analysis. (UQ = Uncertainty Quantification, Lab = With Laboratory Data.).

Model	UQ	Lab	Performance	Remarks
Seq2Seq + GPR Residual (2023) [54]	✓	✗	Error <1.6%	Time-series forecasting using statistical features from charging curves
GPR-LSTM (2023) [124]	✗	✓	RMSPE 2.51% (lab), <3% (real)	GPR reconstruction enables feature extraction from incomplete random charging without complex IC/DV
PSO-VMD-RF-ConvLSTM (2024) [125]	✗	✗	$R^2 \approx 0.98$, MAPE $\approx$ 0.5%	Hybrid model captures temporal–spatial relations in physical charging features
DAE-Transformer (2024) [126]	✗	✓	RE $\approx$ 0.91%	Efficient use of physical knee-point features for fast RUL prognosis
CNN-LSTM (2024) [127]	✗	✗	MAE < 2.13%, RMSE < 2.74%	End-to-end learning using multi-neighboring incomplete charging segments
LC-GDAT Transformer (2025) [128]	✗	✓	SOH error 0.46% (lab), 2.23% (real)	Efficient data compression, dynamic multi-scale feature integration
CNN-Transformer-KAN (2025) [60]	✗	✗	MAE 0.79 Ah, MAPE 0.65%, RMSE 1.02 Ah	Efficient compression of raw charging data into feature maps

Table 10
Thematic evolution of battery state estimation using real-world field data.

Research challenge	Key methodologies	Primary contribution	Representative studies
Adapting to Dynamic Conditions & Noise	Neural Networks (LSTM, GANs)	Capturing transient behaviors and modeling real-world uncertainty for more robust SOC estimation.	Hong et al. 2020 [24], Gu et al. 2023 [133], Jiménez-Bermejo et al. 2025 [134]
Overcoming Lack of Ground Truth & Labeled Data	ML on Operational Data, Semi-/Self-Supervised Learning	Eliminating dependency on lab tests by using cloud data directly; leveraging unlabeled data to improve accuracy with minimal labels.	Li et al. 2024 [114], Yang et al. 2025 [115]
Generalizing Across Diverse Fleets & Behaviors	Fleet-level Analytics, Transfer Learning, Domain Adaptation, Continual Learning	Identifying links between user behavior and SOH; adapting models trained on large fleets to new vehicles for better generalization.	He et al. 2021 [109], Liu et al. 2024 [113]
Predicting Real-World RUL	Hybrid Models (Data-driven + Degradation)	Forecasting long-term degradation based on fleet-specific patterns and real-world charging/driving data.	Zhang et al. 2024 [5], Li et al. 2025 [46]
Enhancing Trust & Practicality	Physics-Informed Neural Networks (PINNs), Federated Learning	Integrating physical laws for more interpretable and robust models; enabling cross-fleet learning while preserving data privacy.	Hofmann et al. 2023 [110], Chen et al. 2025 [108]

Table 11
Comparison of battery fault detection models. Rows highlighted in gray denote studies that publicly released their datasets, which were subsequently utilized for literature-based comparative analysis. (Interp. = Model Interpretability, Early = Early Detection, Fault Data = Fault Data Requirements.) [148].

Model	Interp.	Early	Fault Data	Performance	Remarks
Dynamical Autoencoder (2023) [55]	△	✗	✗	AUROC 88.6%, ADC 0.085 10^4 CNY	EV-scale validation with cost-aware dynamical modeling
DFM-CA-LSTM Autoencoder (2024) [148]	✗	△	✗	AUC 90.73%, F1 score 83.83%	Unsupervised anomaly detection in dynamic frequency domain
GraphSAGE (2025) [85]	△	❍	❍	AUC 97.32%, Accuracy 92.74%, Recall 98.37%, FPR 7.3%	Vehicle-level fault diagnosis using graph-structured EV fleet data

Fig. 18 shows a representative example (DyAD) [55], while Table 11 summarizes multiple models, highlighting their methods and performance. Together, they complement the flowchart (Fig. 17) by linking conceptual taxonomy to practical benchmarking under real-world conditions.

This paradigm enables two powerful diagnostic strategies. The first is providing the aforementioned early warnings, often using advanced, unsupervised deep learning models [55,85,102,149]. The second is using cell-to-cell consistency as a continuous diagnostic metric [68–70]. While analysis of maintenance logs via short-text mining confirms that

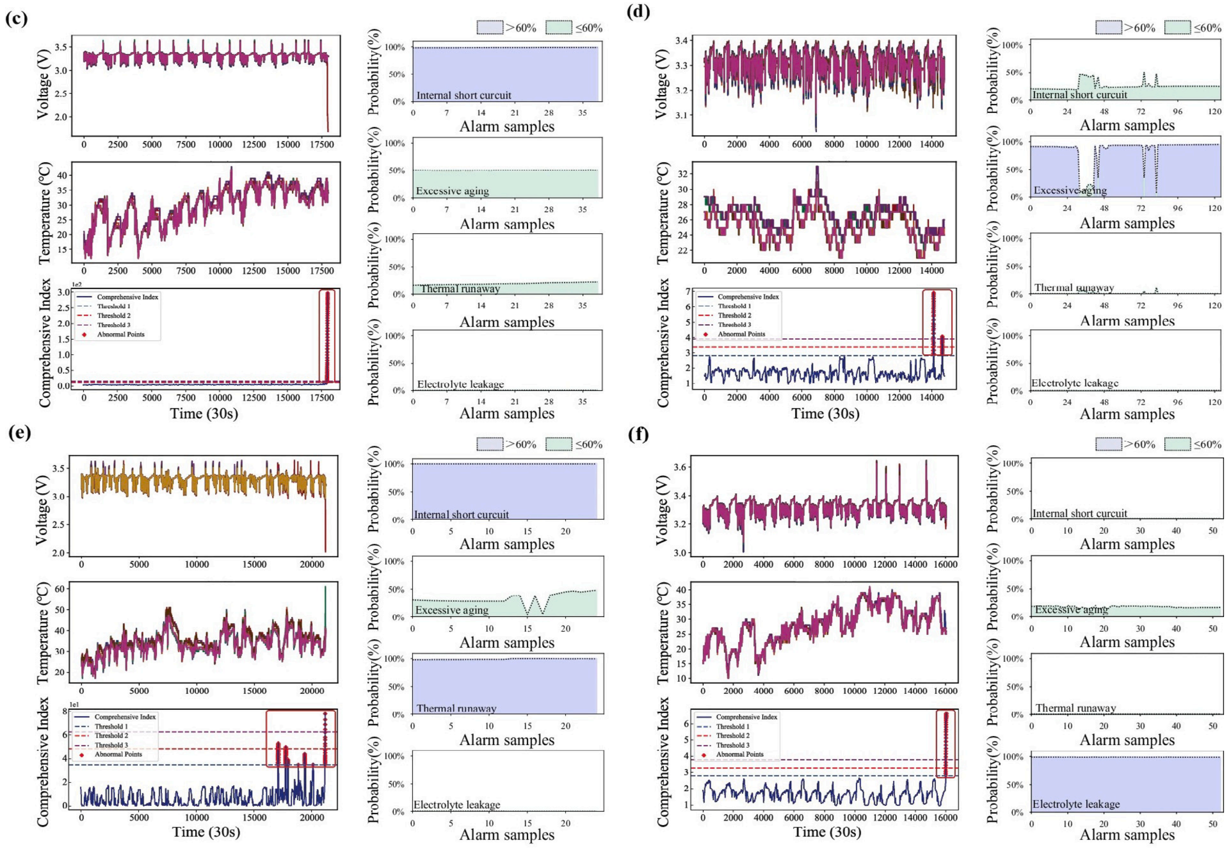


Fig. 16. The left panel illustrates the time-series variations in battery voltage and temperature, while the right panel presents the model-derived results. These signals distinguish between four safety fault types—ISC (internal short circuit), EA (excessive aging), TR (thermal runaway), and EL (electrolyte leakage) [52].

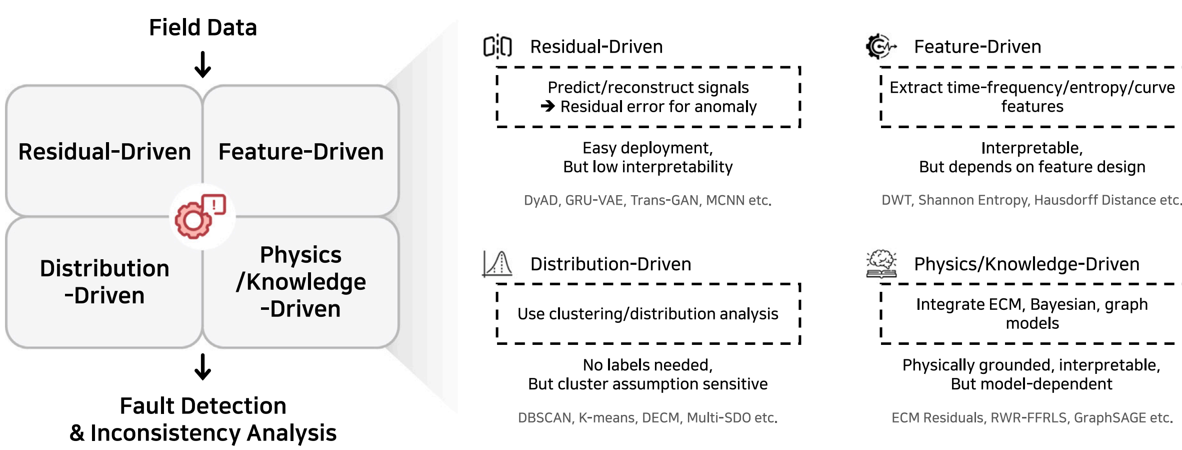


Fig. 17. Systematic flowchart of the battery fault and inconsistency analysis.

"poor cell consistency" is a frequently reported issue as shown in Table 12, field data also demonstrates that there is no direct causal link between inconsistency and failure [150]. A vehicle's actual risk is a complex function of its intrinsic health, the protective limits of the BMS, and the stress imposed by driver behavior [151]. This finding suggests that benchmarking diagnostic algorithms solely on classification accuracy is an oversimplification and can be misleading. The thematic evolution of these fault detection methods is outlined in Table 12.

However, the dominant unsupervised learning paradigm faces significant practical challenges. A primary criticism is the lack of physical interpretability, as many models operate as 'black boxes' unable to link anomalies to electrochemical causes. This is compounded by a fundamental validation problem in verifying the timeliness of "early" warnings with only coarse, vehicle-level fault labels. Furthermore, practical hurdles such as high computational costs, a dependency on refined charging data, and poor generalization often lead to high false alarm rates, hindering deployment. To move forward, the field must address the criticism that many current approaches apply complex AI without a deep understanding of the underlying failure mechanisms [9,150,153]. Detecting subtle, cumulative degradation requires more than just electrical data [21,49,154]. The frontier lies in multi-modal fusion – integrating new sensors like acoustic or gas detectors for direct physical insight – and leveraging unconventional data like maintenance records to ground models in documented, real-world failure modes, thereby creating more robust and interpretable frameworks (see Table 13).

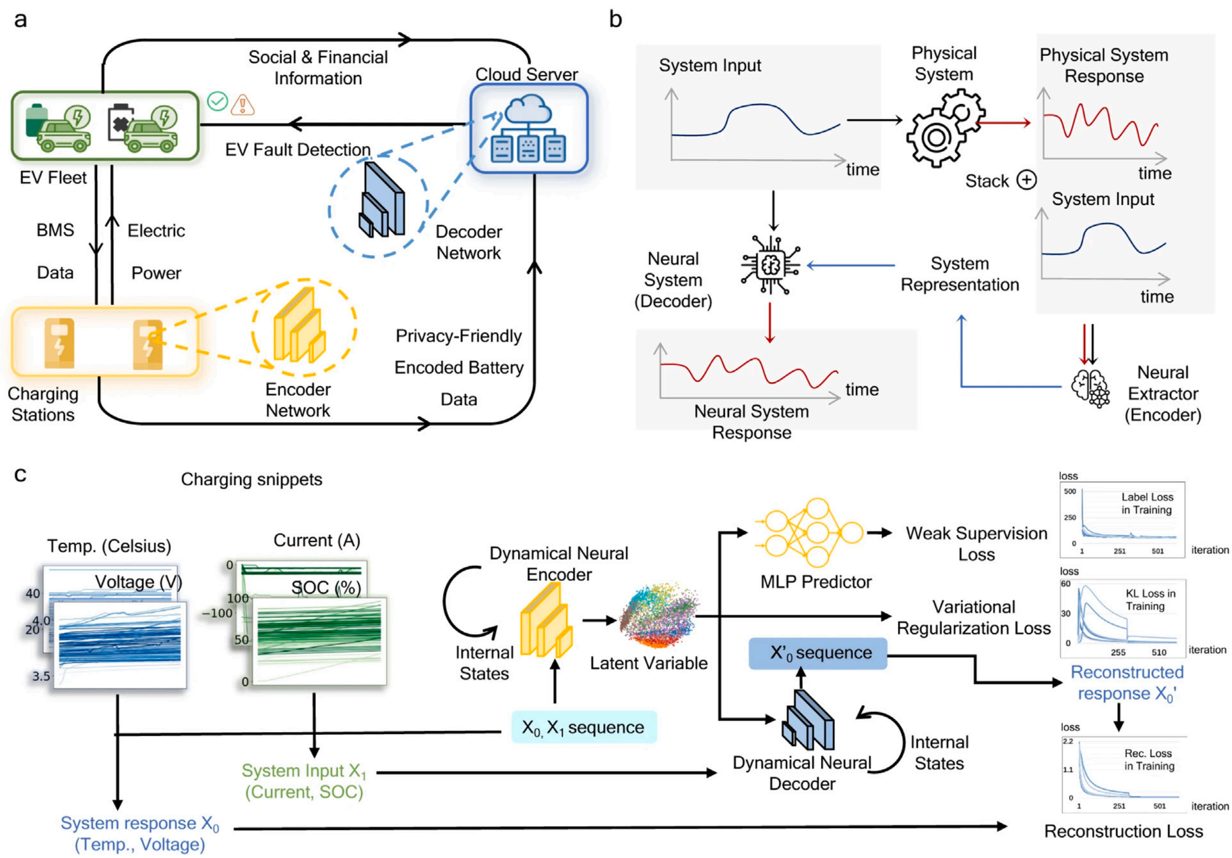


Fig. 18. Fault detection research; dynamical autoencoder for anomaly detection (DyAD) [55].

Table 12
Top 10 key faults and word frequency analysis results [152].

Fault names	Word frequency
poor consistency of battery cells	12,506
open circuit (high voltage)	6979
communication failure between BMS and charger	5781
SOC value is too high	5194
motor controller undervoltage 24 V	4690
over temperature fault of charging base	4522
overcurrent (high pressure)	3925
heating or water-cooling relay failure	3574
battery cell voltage is too high	2502

4.3. Energy consumption and utilization analysis

Analysis of real-world field data fundamentally deconstructs the standardized view of energy consumption provided by laboratory drive cycles [157]. This research reveals that actual user behavior, such as aggressive versus calm driving, and environmental conditions, especially ambient temperature, are often the dominant factors influencing real-world efficiency and battery utilization [56]. For instance, large-scale fleet studies [45,158] have quantified this impact, showing that seasonal temperature shifts can alter average energy consumption by as much as 21% even when monthly driving demand remains constant (Fig. 19(a)). Furthermore, the analysis of field data demonstrates that the effectiveness of key technologies like regenerative braking is highly context-dependent, with its contribution to energy recovery being significantly greater in urban stop-and-go traffic than on highways. This body of work reframes vehicle efficiency, moving it from a static property to a dynamic, environment-dependent state.

The most transformative application of analyzing aggregated data at the fleet level is its ability to inform system-wide optimization and macro-scale decision-making. By leveraging data from thousands of vehicles, researchers can devise data-driven strategies for charging infrastructure planning [44], where station placement is optimized based on real-world demand patterns and charging times can be more accurately predicted (Fig. 19(b)). The analysis also highlights how utilization profiles differ across various applications – such as e-buses, e-boats, and different commercial fleets – thus bridging the gap between vehicle engineering and urban energy system design [57] (Fig. 19(c)).

Ultimately, the analysis of field data enables a shift from a static, manufacturer-stated driving range to a dynamic and personalized range prediction [159–161]. As demonstrated in multiple studies, models trained on this data can provide forecasts tailored to current temperature, a driver's recent behavior, and the intended route type. This capability to provide realistic, context-aware range estimates is critical for overcoming "range anxiety" and increasing consumer confidence in EV technology. The thematic evolution of EV energy consumption and utilization is outlined in Table 14.

4.4. Limitations across research trends

Recurring challenges remain evident throughout this review. At a fundamental level, uncertain ground-truth labels, corrupted diagnostic signatures, heterogeneous data formats, and privacy constraints – outlined in Section 3 – persist as structural obstacles. Recent research trends further highlight domain-specific limitations: state and lifetime estimation suffers from missing capacity benchmarks and limited generalization across fleets; fault detection is hampered by scarce labeled failures, weak physical interpretability, and frequent false alarms; and energy consumption studies, while strong at the fleet level, struggle to deliver reliable individualized predictions under behavioral and environmental variability. These limitations collectively emphasize the

Table 13
Thematic evolution of fault detection methods.

Research phase	Key methodologies	Primary contribution	Representative studies
Early Indicators (pre-2018)	Statistical & Entropy-Based Analysis Clustering	Identifying early fault indicators like voltage randomness and entropy changes in bulk data.	Zheng et al. 2013 [155], Hong et al. 2017 [20]
Machine Learning for Temporal Analysis (2019–2022)	Time-series NNs Hybrid Models (ML + ECM) Autoencoders	Capturing time-series dependencies to predict fault evolution and linking real-world data to physical models.	Hong et al. 2019 [19], Li et al. 2021 [156], Jiang et al. 2021 [39]
Advanced DL for Robustness & Context (2023–2024)	Transformers, GNNs Multi-modal Fusion Adaptive thresholds	Achieved high accuracy and generalization on diverse, heterogeneous data.	Zhao et al. 2023 [139], Zhang et al. 2023 [55], Tang et al. 2024 [102]
Real-Time & Model-Constrained Diagnosis (2025+)	Model-Constrained Neural Networks (MCNN), Distribution-calibrated models	Attained high diagnostic accuracy with low false alarms under uncertain conditions.	Cao et al. 2025 [52], Zhang et al, 2025 [85]

Table 14
Thematic evolution of EV energy consumption and utilization analysis.

Research theme	Key factors analyzed	Primary contribution	Representative studies
Real-World Efficiency	Driver Behavior, Weather, Route Type	Vehicle efficiency varies dynamically due to real-world conditions	Zhang et al. 2020 [157], Shen et al. 2023 [160]
Fleet-Level System Optimization	Fleet Composition, Charging Demand, Urban Mobility	Fleet data drives optimization of charging infrastructure and grid electrification	Zhao et al. 2021 [45], Tepe et al. 2023 [57], Rücker et al. 2024 [56]
Personalized Predictive Analytics	Driving Data, Route Context, Traffic Insights	Field data supports dynamic, personalized range predictions over static estimates	Bustos et al. 2025 [161], Zhu et al. 2025 [162], Wu et al. 2025 [44]

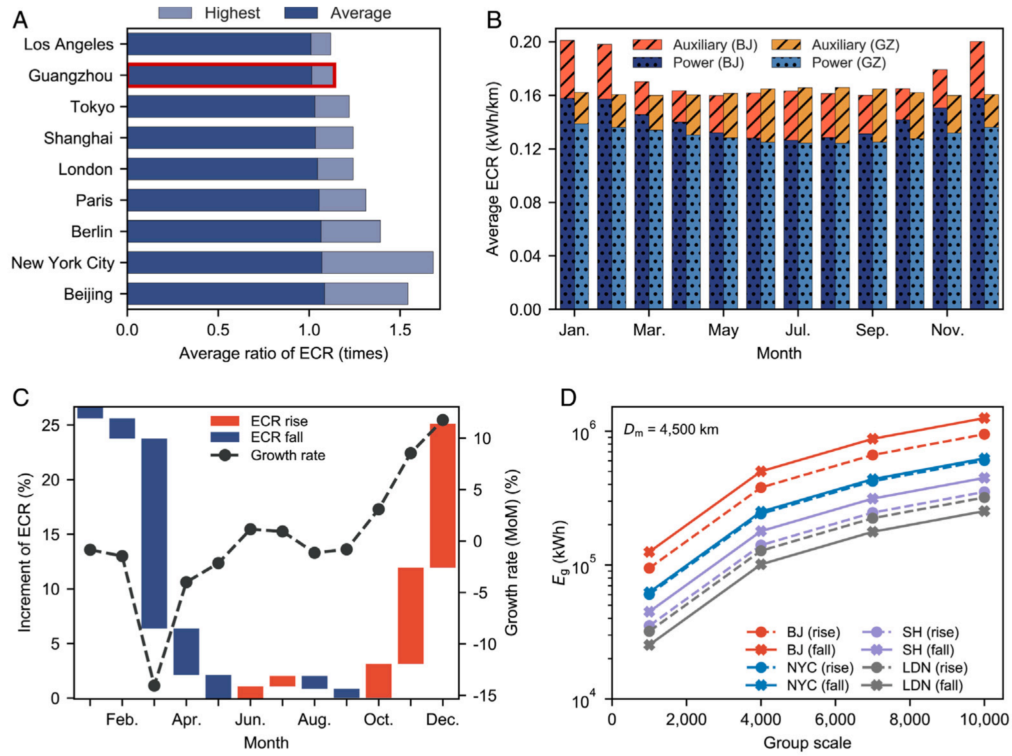
need for richer and standardized datasets, integrated lab–field frameworks, and interpretable models to enable a transition from descriptive insights toward predictive and prescriptive field-data-driven battery management.

5. Future outlook and opportunities

This review first explores the potential of EV field datasets (Section 2), then discusses the complex challenges associated with their processing and analysis (Section 3), and finally presents the tangible applications and insights they unlock (Section 4). The field now stands at a critical inflection point, where recent discoveries are not merely refining existing knowledge but are fundamentally reshaping it. Groundbreaking research from Geslin et al. [11] has revealed that real-world dynamic driving can enhance battery lifetime by up to 38% over conventional static lab tests, a finding explained by the beneficial effects of low-frequency current pulses inherent in such cycles. This is powerfully corroborated by research from Schreiber et al. [8], which shows that much of the degradation in lab tests is recoverable "apparent aging", not permanent damage. Static tests, therefore, do not just simplify reality—they can distort it, as up to 52% of capacity loss can be recovered after a rest period. These revelations confirm that the future of battery innovation lies not in the controlled vacuum of the lab, but in deciphering the complex, dynamic language of the field.

This paradigm shift unlocks transformative opportunities, redirecting industry focus from passive diagnostics to proactive, intelligent lifecycle management. The insight that real-world conditions are often less damaging than previously predicted creates a compelling business case for second-life battery applications, converting retired units from liabilities into valuable assets. It provides the foundation for developing dynamic, health-aware control strategies, such as intelligent charging protocols and adaptive usage recommendations that actively extend a battery's primary service life. Ultimately, batteries become data-rich assets whose value can be actively managed and optimized, aligning with industry-wide standardization efforts dependent on reliable battery state information.

Realizing this vision requires fundamental advances in both vehicle architecture and modeling methodologies. The vehicle's electrical/electronic (E/E) architecture is shifting from a distributed design with numerous Electronic Control Units (ECUs) to a centralized framework powered by robust High-Performance Computers (HPCs), providing the necessary computational foundation. Concurrently, the modeling techniques are advancing beyond the limitations of purely data-driven "black box" approaches toward Physics-Informed Machine Learning (PIML). By embedding core electrochemical principles, PIML becomes not only highly accurate but also interpretable and robust in out-of-distribution scenarios—a critical requirement for establishing trust in AI-driven battery management. At the same time, practical deployment at the edge requires models to be lightweight, pointing toward embedded AI and surrogate-based approaches for real-time inference under tight energy budgets. Beyond such model compression, recent advances in Physical Neural Networks (PNNs) [163] suggest a longer-term direction, where analogue physical systems themselves can serve as efficient computing substrates with drastically reduced energy demands. The deployment of these models is facilitated by industry-wide standards like Service-Oriented Vehicle Diagnostics (SOVD) and the Extended Vehicle (ExVe, ISO 20078), which provide the standardized interfaces for these intelligent systems to operate cohesively across the ecosystem, as illustrated in Fig. 20.



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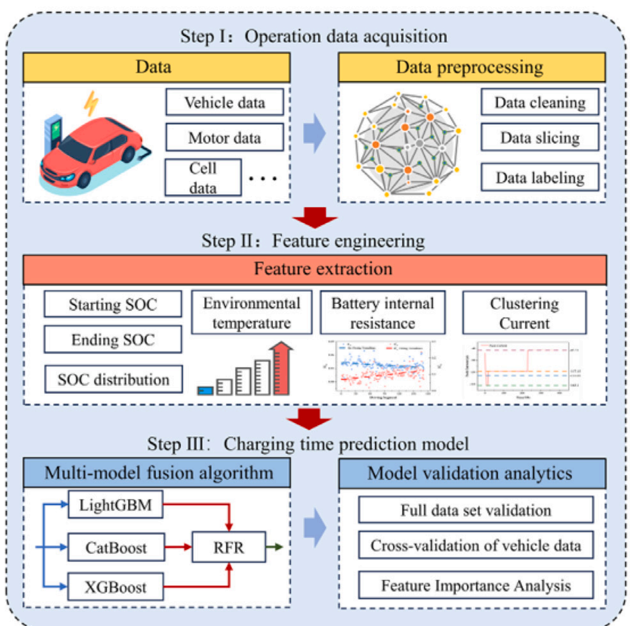

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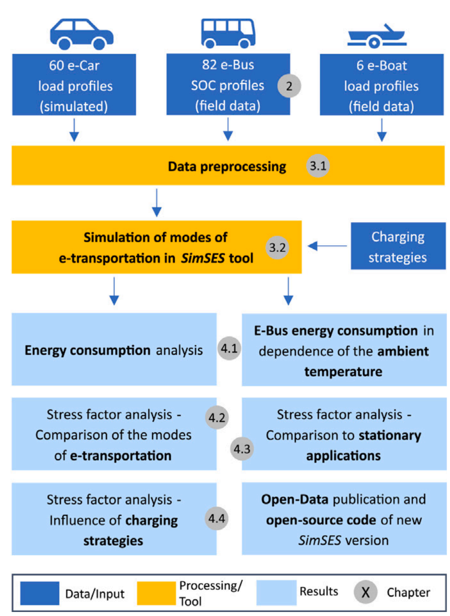

(c)

Fig. 19. Energy consumption and utilization analysis research; (a) regional and monthly changes in EV energy consumption [45], (b) structure of the fusion model for charging time prediction [44], and (c) lithium-ion battery utilization in various modes of e-transportation [57].

This convergence of rich field data, advanced computing, and trustworthy AI points toward a singular, transformative objective: the creation of a generalized battery foundation model [164–166]. Inspired by large language models, this paradigm involves pre-training on massive, multi-chemistry datasets, enabling rapid fine-tuning for specialized tasks from diagnostics to advanced control. It provides the basis for perpetually learning digital twins for every battery—dynamic models continuously updated by real-world data, representing the ultimate realization of the concepts presented in this paper. In this envisioned future, a battery's lifecycle is no longer merely predicted, but actively managed and optimized, thereby maximizing operational value and contributing significantly to a sustainable energy ecosystem. Looking

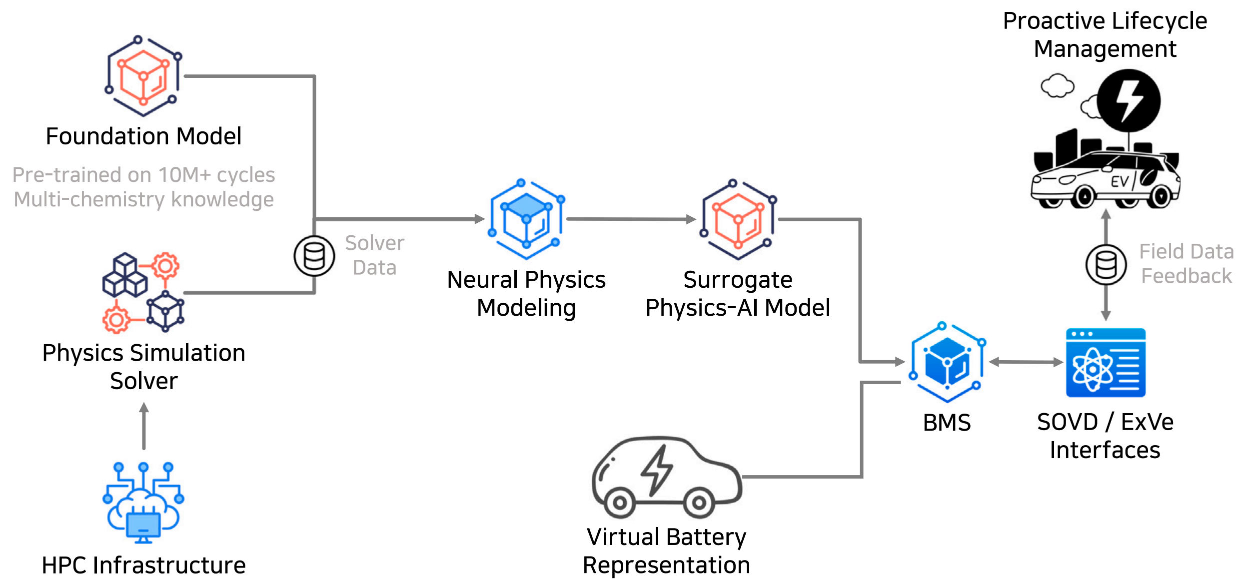

Fig. 20. Conceptual framework for a field-data-driven battery foundation model.

ahead, realizing this vision will require a forward-looking research agenda that prioritizes opportunities such as data format standardization, privacy-preserving analytics, and multi-modal fusion with non-electrical signals, thereby establishing a long-term roadmap for the community.

CRediT authorship contribution statement

Kyunghyun Kim: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kyeongeun Cho:** Investigation. **Kwangho Lee:** Investigation. **Junyoung Yoon:** Investigation. **Jung-Il Choi:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini to polish the writing. After using this tool/service, all authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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